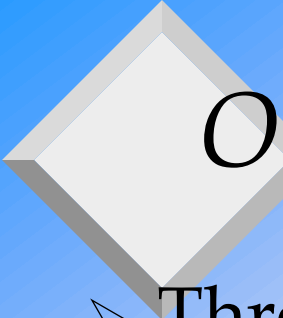




Data Warehouse Architectures





Objectives of Today's Lecture

- Three-Layer architecture
- Enterprise data model
- Meta data
- Data part in three-layer architecture
- Status Vs Event data
- Transient Vs Periodic data
- Extract and Types of Extract
- Loading Two modes



Three-Layer Data Architecture

- Three important terms
 1. Operational data
 2. Reconciled data
 3. Derived data



Three-Layer Data Architecture Contd..

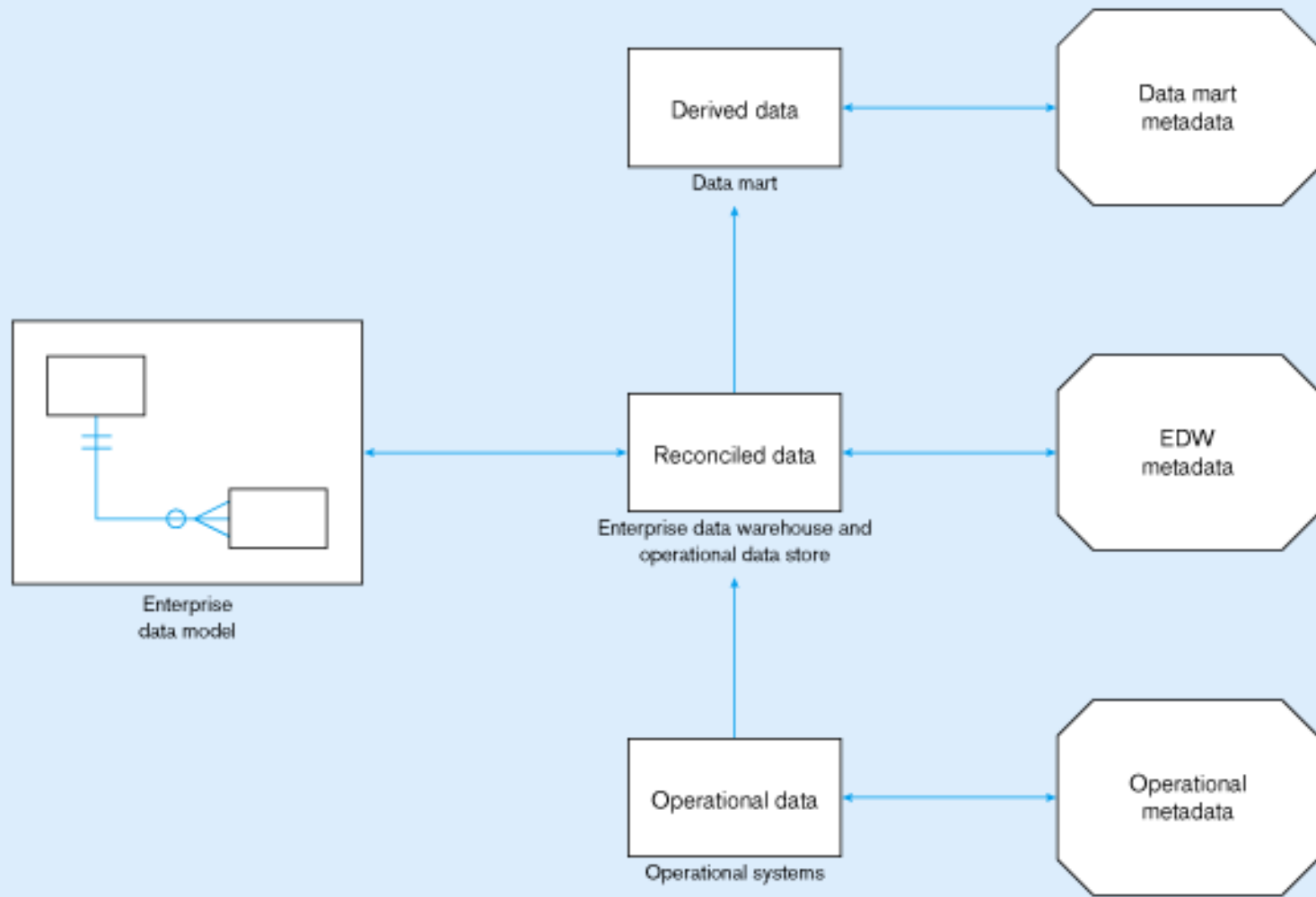
- Operational data ?
- Reconciled data
 - Is detailed, current data intended to be the single, authoritative source for all decision support applications
- Derived data
 - Data that have been selected, formatted, and aggregated for end user decision support application



Three-Layer Data Architecture Contd..

- Two components plays important role in this architecture :
 - Enterprise data model
 - Meta data

Three-layer architecture





Role of Enterprise Data Model

- It presents total picture explaining the data required by an organization
- It control the phased evolution of DWH
- It takes too long to develop the enterprise data model in one step and dynamic need for decision making will change before the warehouse is built



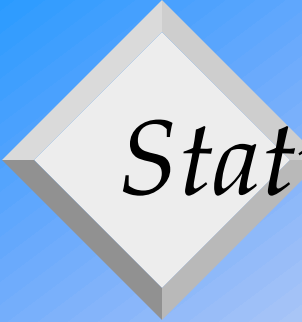
Role Of Meta Data

- Meta data are data that describes the properties or or characteristics of other data
- Operational meta data
 - Describes data in various operational systems
- EDWH meta data
 - Derived from EDWH
 - Describes the reconciled data layer as well as the rules rules for transforming operational data into reconciled data
- Data mart Meta Data
 - Describes the derived data layer and the rules for transforming reconciled data to derived data



Data part in three-layer architecture

- Operational Data?
- Reconciled data
 - Are detailed, current data intended to be the single, authoritative source for all decision support applications
- Derived data
 - Data that have been selected, formatted, and aggregated for END USER decision support applications



Status Vs Event data

- Status data
 - Before and after image of data
- Event data
 - Data on which action/event is performed
- Event
 - A database action (create, update, or delete) that results from a transaction.
 - A transaction may lead to one or more events like in case of :
 - Withdrawal
 - transfer
- In practice most of the data stored in DB is status data
- Both data are typically stored in DB logs for backup and recovery



Transient Vs Periodic data

- Transient data
 - Data in which changes to existing records are written over previous records
 - It destroys the previous data
- Periodic data
 - Data that are never physically altered or deleted, once they have been added to the store.



Example of Periodic and Transient Data



Data reconciliation

- It can be visualized as a process consisting of four steps
 - Capture
 - Scrub
 - Transformation
 - loading



Extract

- Capturing the relevant data from the source files and DBs used to fill the EDW
- Types of Extract
 - Static extract
 - Incremental extract



Extract Contd..

➤ Static extract

- A method of capturing a snapshot of the required source data at a point in time
- Used to fill DWH initially

➤ Incremental extract

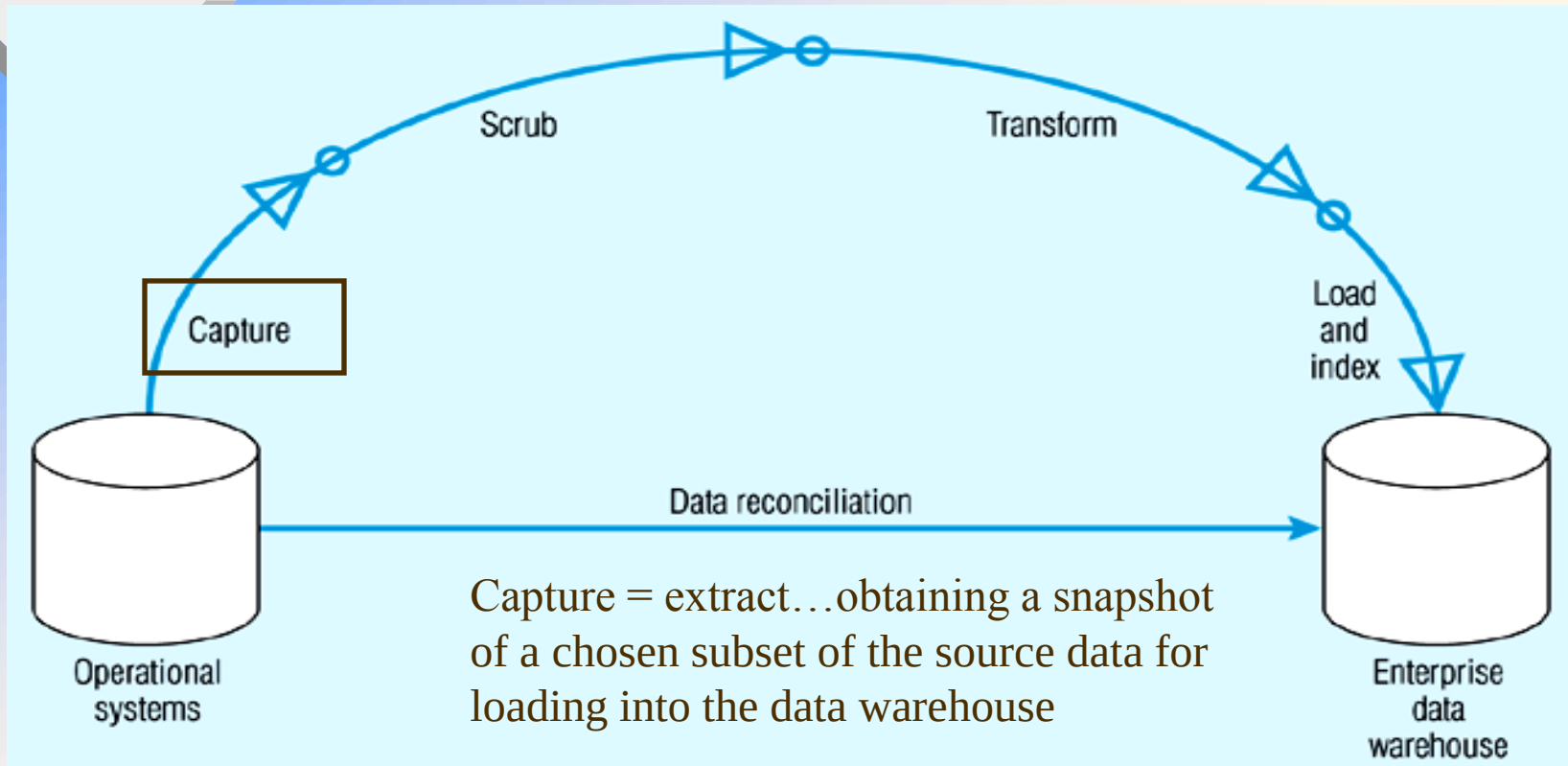
- A method of capturing only the changes that have occurred in the source data since the last capture
- Used for ongoing warehouse maintenance



Data Scrubbing / cleansing

- A technique using pattern recognition and other artificial intelligence techniques to upgrade the quality of raw data before transforming and moving the data to the warehouse
- Which data needs to be scrubbed
 - Misspelled names and addresses
 - Impossible or erroneous dates of birth
 - Fields used for the purpose for which it was never intended
 - Missing data
 - Duplicate data
 - Mismatched addresses or area codes

Steps in data reconciliation



Static extract = capturing a snapshot of the source data at a point in time

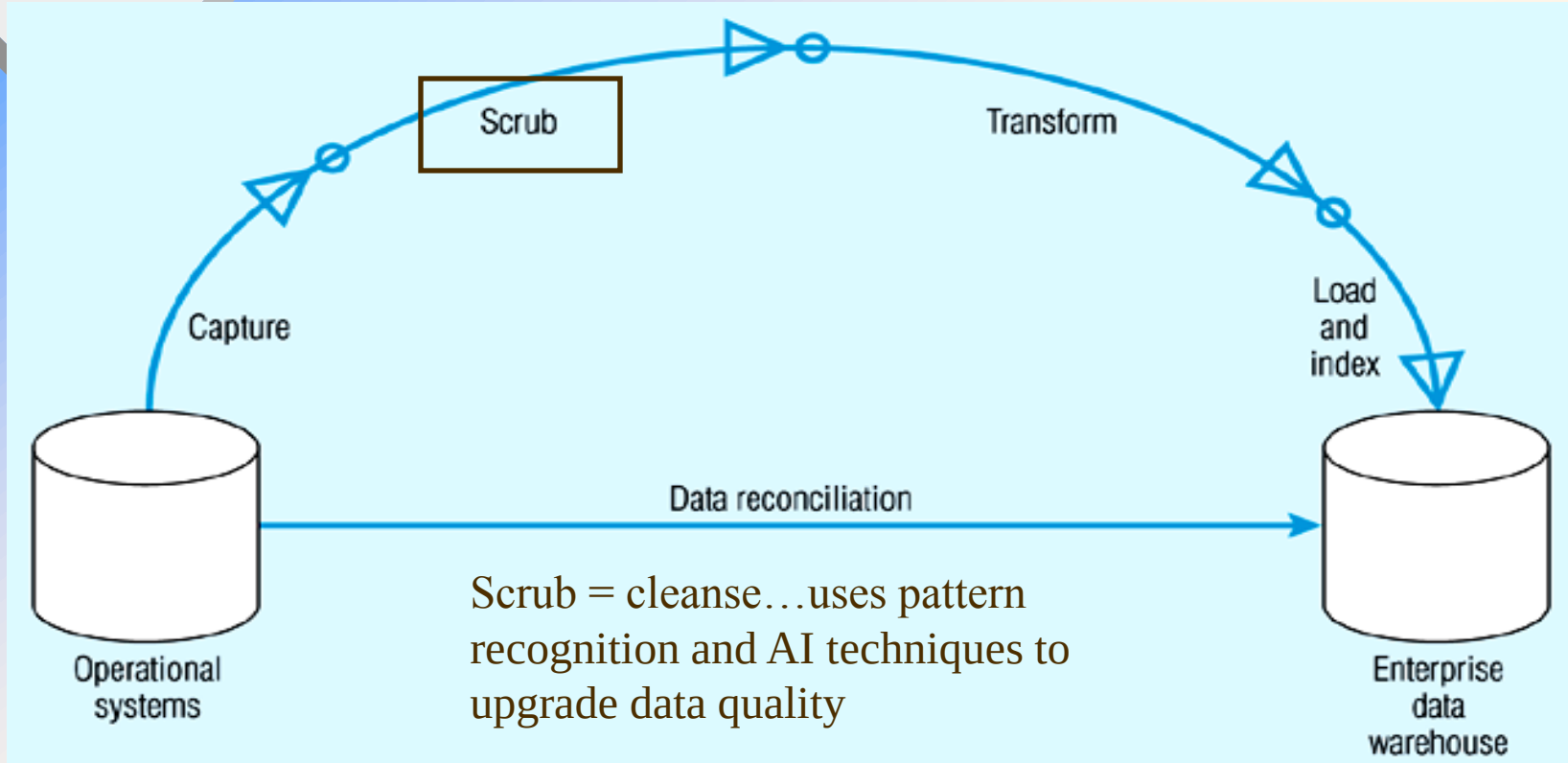
Incremental extract = capturing changes that have occurred since the last static extract



Data Scrubbing / cleansing Contd..

- Bandwidth of Cleansing depends upon quality of data
 - Higher the quality less effort will be needed for cleansing and vice versa
- Common cleansing tasks are
 - Decoding data to make them understandable for DWH applications
 - Reformatting and changing data types
 - Converting between different measuring units
 - Finding missing data to complete the batch of data necessary for subsequent loading

Steps in data reconciliation (continued)



Fixing errors: misspellings, erroneous dates, incorrect field usage, mismatched addresses, missing data, duplicate data, inconsistencies

Also: decoding, reformatting, time stamping, conversion, key generation, merging, error detection/logging, locating missing data



Load and Index

- Two modes
 - Refresh mode
 - Update mode



Load and Index Contd..

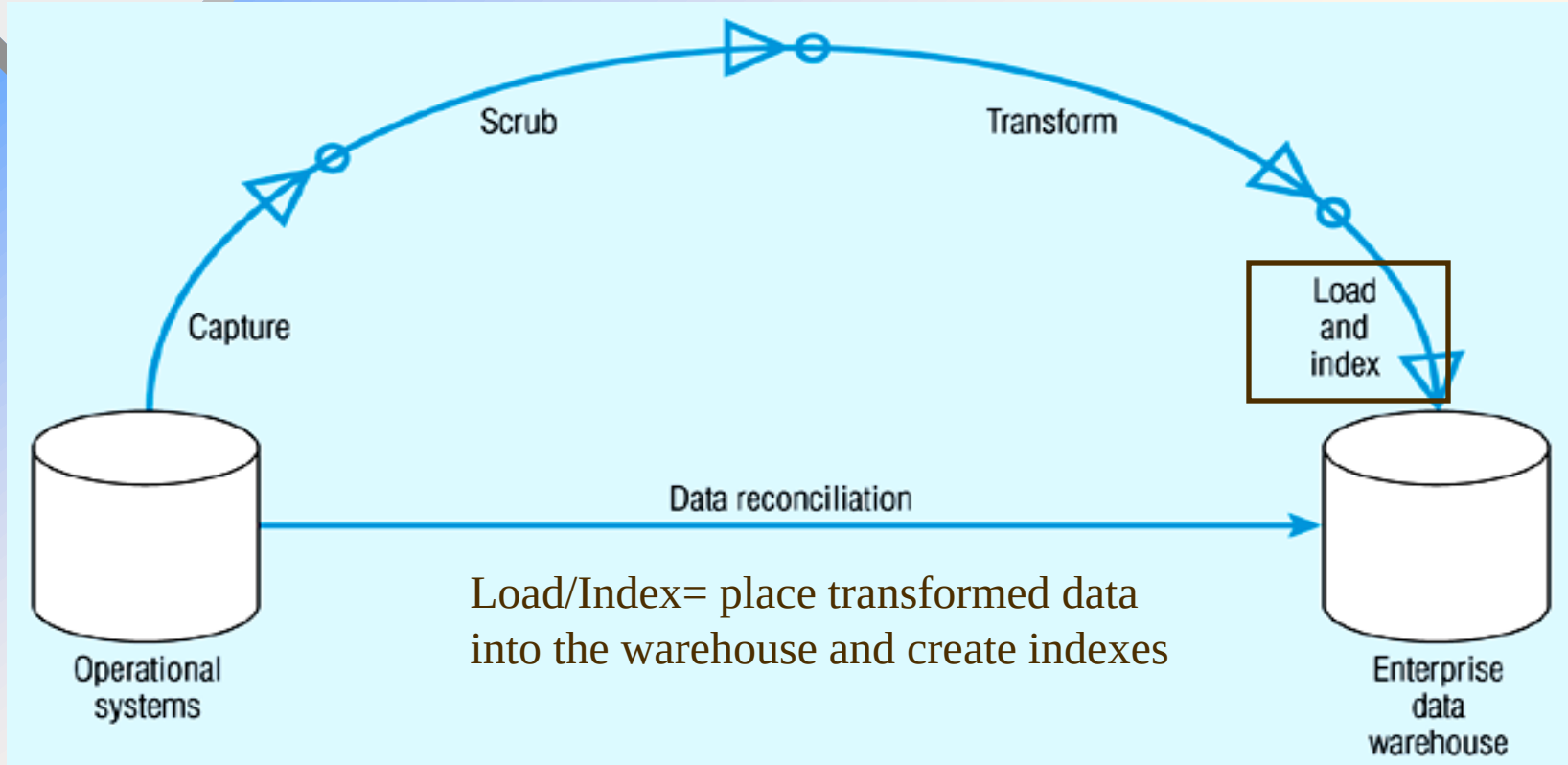
➤ Refresh mode

- An approach to fill the DWH that employs bulk rewriting of the target data at periodic intervals
- Replaces the previous contents
- Less popular
- Good for filling DWH initially
- Used in conjunction with static data capture

➤ Update mode

- An approach in which only changes in the source data are written to the DWH
- New records are written without overwriting previous record
- Used in connection with incremental data capture

Steps in data reconciliation (continued)



Refresh mode: bulk rewriting of target data at periodic intervals

Update mode: only changes in source data are written to data warehouse



Thank You Very Much